



Scleral complications of plaque brachytherapy for pterygium and for uveal melanoma

The authors read with interest Jabbarli et al.'s article on the incidence and risk factors for radiation-induced scleral necrosis (RISN) following brachytherapy for uveal melanoma (UM).¹ They would like to comment further on the historical incidence of RISN, other potential sequelae and their relevant management. In Australia, there is a particular interest in RISN due to the historical incidence of RISN related to beta radiation plaque brachytherapy formerly used as an adjunct to pterygium excisional surgery. Accordingly, the authors would like to make two observations.

Firstly, similar correlations between historical data of RISN associated with pterygium excision and brachytherapy allow us to predict potential sequelae associated with brachytherapy used for UM. However, it must be noted that the radioactive materials used in brachytherapy for UM differ from those used following pterygium excision. Jabbarli et al.'s case series focuses on patients who had received either Ruthenium-106 (¹⁰⁶Ru) or bi-nuclide radioactive plaques of Ruthenium-106 (¹⁰⁶Ru) and ¹²⁵Iodine (¹²⁵I) seeds.¹ In contrast, beta radiation for pterygia treatment traditionally uses Strontium-90 (⁹⁰Sr).²

In clinical ophthalmology, the use of ⁹⁰Sr was first described in 1950 by Friedell in the treatment of superficial benign tumours such as papillomas of the lids and conjunctiva, as well as vascular abnormalities such as haemangiomas.² Although both ¹⁰⁶Ru and ⁹⁰Sr primarily emit beta radiation, such that they have similar effects on ocular tissue, they have different levels of ocular penetration, with ⁹⁰Sr penetrating less than ¹⁰⁶Ru.²

⁹⁰Sr brachytherapy reduces the rate of tumour recurrence when used in conjunction with primary surgical excision of pterygium. It gained popularity in the 1970s in Australia due to the high prevalence and recurrence of pterygia. In 1980, Tarr and Constable performed a retrospective study on 63 eyes (57 patients) who had late complications following beta radiation in the management of pterygia in Western Australia.³ Scleral ulceration occurred in 52 eyes of 46 patients, with 31 of those eyes suffering deep scleral ulceration (ulceration at a depth of two-thirds or greater). These findings were substantiated in a 10-year retrospective case series out of Queensland by Mackenzie's group, which looked at 764 eyes (585 patients) who had undergone beta radiation in the management of pterygia.⁴ They noted a RISN incidence of 4.5%.⁴ This result is comparable with the 2.9% incidence

rate of RISN found by Jabbarli et al.¹ These two studies prompted Australian ophthalmologists to reassess the need for beta radiation, which was subsequently replaced by pterygium excision and conjunctival autografting.

It is significant that the data from Mackenzie's group⁴ demonstrate the comparable rates of RISN following both beta radiation for pterygium removal and brachytherapy for UM. Unsurprisingly, this form of beta radiation produces similar scleral pathology, even when treating different disorders.

Secondly, previous studies addressing RISN-related endophthalmitis may be useful in guiding treatment of RISN in UM patients. Whilst Jabbarli et al. did not comment on the rate of endophthalmitis associated with scleral necrosis, Tarr and Constable reported a correlation. They outlined four cases of what the authors would now label as RISN-associated endophthalmitis caused by particularly virulent strains of *Pseudomonas aeruginosa*.³ Following the failure of traditional treatments of RISN-associated infections using topical and subconjunctival fortified antibiotics, continuous infusions of tobramycin into the superior fornix were required to treat the infection successfully. This is an example of subpalpebral antibiotic lavage (SAL), recently reported and reviewed by the authors. SAL is a cost-effective, safe and therapeutic second line treatment, but despite its clear advantages, the literature reveals an apparent underutilisation of this technique.⁵

The authors invite Jabbarli et al. to comment on the management and outcomes of their patients who developed RISN, and particularly the rate of RISN-associated endophthalmitis and globe perforation.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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Scleral complications of plaque brachytherapy for pterygium and for uveal melanoma: Response

We would like to thank Dr Xiradis and colleagues for the comments¹ to our paper about the radiation induced scleral necrosis (RISN) after brachytherapy of uveal melanoma (UM).² The authors draw a parallel between data of scleral necrosis (SN) associated with pterygium excision and adjunctive brachytherapy with Strontium-90 (⁹⁰Sr) and RISN following brachytherapy with Ruthenium-106 (¹⁰⁶Ru) and Ruthenium-106 - Iodine-125 (¹⁰⁶Ru - ²⁵I) plaques in UM.

Pterygium excision is the most common ocular surgery related to surgical induced SN.³ Inhibition of repopulation of endothelial cells due to brachytherapy minimises the local recurrence rate.⁴ Taking into consideration that the cases of SN were also reported after bare sclera technique without any adjuvant therapy one can presume that irradiation is just one of the contributing factors.⁴

Though both isotopes ¹⁰⁶Ru and ⁹⁰Sr are beta emitters, there is an important aspect that has to be considered for comparing biological aspects of the eye plaques based the dose rate of each isotope. ⁹⁰Sr plaques have a dose at of ~1–2 Gy/min whereas ¹⁰⁶Ru-plaques deliver a dose of 0.08–0.17 Gy/min.⁵ For ⁹⁰Sr plaques, the rules of high dose rate (HDR) brachytherapy apply, whereas for ¹⁰⁶Ru plaques, those of low dose rate (LDR) apply. In other words, at the same applied dose in a single fraction, there will be more toxicity for HDR. The fractionation therefore is an important aspect for HDR brachytherapy.

Ali et al. reported various postoperative irradiation schemes in pterygium patients with doses from 10 to 60 Gy divided in 1–6 fractions for 1–6 weeks starting within 3 days postoperatively.⁴ Due to a rapid dose fall-off, much less dose reaches sclera with ⁹⁰Sr. In contrast, significantly higher scleral doses are being applied during brachytherapy of UM with ¹⁰⁶Ru plaque (≥700 Gy, as it has been shown in our series²). Therefore, comparable rates of SN despite disparity on applied scleral doses might be related to differences on relative biological effectiveness of ⁹⁰Sr and ¹⁰⁶Ru.

Infection is another relevant risk factor associated with SN after pterygium surgery. So, Doshi et al. reported >70% infection rate in 203 SN cases after pterygium surgery.³ However, a link between perioperative infection and RISN is rather uncommon after brachytherapy for UM. Therefore, there were no cases with RISN and infectious scleritis in our series.²

Another points highlighted by Xiradis et al.¹ are the management and outcome of RISN, including the risk of RISN related endophthalmitis. Severe thinning of the exposed irradiated sclera might increase the risk of endophthalmitis in patients with RISN, eventually due to inapparent microperforation.³ In our cohort, there were no cases with RISN-related endophthalmitis.² As to conservative and surgical management of RISN, we will address this question in a separate publication.

In conclusion, we greatly appreciate the comments made by the authors and thank for their interest in our paper.